Subset-sum landscapes under deformed and random measures: the biased invariant $\Gamma^{(q)}$ and unconditional rigidity

(working title — draft, round 107)

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Abstract

Papers 1–3 of this series fixed both the sequence and the measure on its subsets. Here we move each. Replacing the uniform measure by a Bernoulli(q) product measure deforms the gap series into an explicit family $\Gamma^{(q)}$, and we prove that the fair coin minimises it for every sequence, with equality only at $q = \frac{1}{2}$; the argument is termwise and is machine-checked in Lean 4. We derive the corresponding transfer function $\Phi^{(q)}$ and show that the bias splits paper 3's symmetric pair of strata in the ratio $q : (1-q)$, so that the biased landscape responds to a shift of target at first order; the resulting slope is a point prediction with no free parameter, and it is confirmed against the natural alternative to within 0.4%. Replacing the primes by a random odd sequence, we prove that the extremal modulus of the minor arcs moves from 6 to 4 and the geometric rate improves from $\sqrt{3}/2$ to $1/\sqrt{2}$, so the primes are the harder case; the remaining half of that statement is a second-moment argument that carries no ineffective constant. Two questions inherited from paper 3 are also settled or sharpened. Every claim is tagged with its status, and the numerical work is reproducible from logged scripts.

1 Introduction

Papers 1–3 fixed two things at once: the sequence A, and the measure on its subsets. Every statement of the trilogy is about a specific A under the uniform measure. This paper moves each of them, one at a time, and the two moves turn out to be worth making for different reasons.

Bias. The classification theorem of paper 1 says which elements lie on which side of a target. It says nothing about how configurations are weighted, so the entire combinatorial layer survives verbatim when the uniform measure is replaced by the Bernoulli(q) product measure, and the gap series deforms into an explicit one-parameter family $\Gamma^{(q)}$ (§2). The answer has a shape worth putting in the introduction: **the fair coin gives the least rugged landscape**, for every sequence, by a termwise convexity argument that is machine-checked (Theorem 2.5, Remark 2.4). The transfer function deforms too, and here the deformation is not cosmetic: paper 3's cosh splits into two unequal exponentials with offsets in the ratio $q : (1-q)$, so the biased landscape responds to a shift of target at *first* order rather than second (§3). Rigidity off the fair coin is a tilt, not a plateau.

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Randomness. For a random odd sequence the minor-arc input that cost paper 2 fifty rounds is replaced by a second-moment bound, so §4 is expected to carry *no ineffective constant at all* — the first such statement in the series. What we did not expect is that the random case is also *easier*: the extremal modulus moves from 6 to 4 and the geometric rate improves from $\sqrt{3}/2$ to $1/\sqrt{2}$ (Theorem 4.3). The reason is a one-line arithmetic fact — a set of primes contains at most one element $\equiv 3 \pmod 6$, a random odd set contains a third of them — and its consequence is that paper 2's $\sqrt{3}/2$ is a property of the primes, not a limitation of the method. *The primes are the hard case.*

A methodological remark that this paper earned twice. Deformations are not only a way to generalise; they are a way to *see*. **A deformation that breaks a symmetry makes measurable everything the symmetry was hiding.** Constructing $\Gamma^{(q)}$ and $\Phi^{(q)}$ audited paper 3 for free, twice. First, the c_d fork: asking how the omitted factor behaves as k grows, rather than how large it is at one k, closed a question paper 3 had left open (§5). Second, and more sharply, the sign of λ: paper 3's Φ is an *even* function of λ, so no observable in that paper depends on the sign, and a definition that carried the wrong one survived three papers and an external referee. Splitting the cosh made the sign observable in a single measurement. We record the principle because it is a design rule, not an anecdote: when a quantity cannot be checked, deform the problem until it can.

Verification, and the genre. Every numbered claim carries a status tag: proved, Lean-verified, experimentally confirmed with the range stated, or conjectured. The scripts that produce every number in this paper write logs beside themselves and are part of the artefact. This is now a recognisable genre — machine-assisted mathematics shipped with formal certificates, of which *Ten advances in mathematics and theoretical computer science* [1] and its accompanying Lean 4 repository are the visible precedent. The direction differs: that work attacks named open problems, whereas the present series builds a small theory from its definitions upward, so the certificates here anchor *definitions and structural lemmas* rather than headline resolutions. The verification ethos is the same one, and we take it as read rather than arguing for it.

2 The biased invariant

Let P_q be the product measure under which each $a_i \in A$ is present independently with probability $q \in (0,1)$, and write $\mathrm{lm}_q(n)$, $\deg_q(n)$ for the P_q-measures of the sets of strict local minima and of ground states at the target n. The target is taken at the q-tilted centre $n_q = \lfloor qT \rceil$.

Definition 2.1 (biased gap series)**.**

$$\Gamma^{(q)}(A) \;=\; 1 + \sum_{d\geq 1}\Big[q^{N_d} + (1-q)^{N_d}\Big].$$

At $q = \frac{1}{2}$ every bracket is $2 \cdot 2^{-N_d}$ and $\Gamma^{(1/2)}$ is the window series W_D of paper 1, hence $\Gamma(A) + (2D+1)2^{-k}$.

Remark 2.2 (why the classification survives the deformation)*.* Paper 1's classification is measure-free: at error d, an overshooting configuration is a strict local minimum exactly when every $a \leq 2d$ is *excluded*, and an undershooting one exactly when every $a \leq 2d$ is *included*. Under P_q those two events have probability $(1-q)^{N_d}$ and q^{N_d}, which is Definition 2.1. Writing

$r^q_B(m) = \sum_{U\subseteq B,\ \sigma(U)=m} q^{|U|}(1-q)^{|B|-|U|}$ and factorising P_q across I_d and B_d,

$$\mathrm{over}_q = (1-q)^{N_d} r^q_{B_d}(n+d), \quad \mathrm{under}_q = q^{N_d} r^q_{B_d}(n-d-\sigma_d), \quad \deg_q = \sum_{J\subseteq I_d} q^{|J|}(1-q)^{N_d-|J|} r^q_{B_d}(n-\sigma(J)),$$

and the binomial sum over J is exactly 1, so a flat r^q gives the d-th term of Definition 2.1. [derived; the assignment verified per stratum, see Remark 2.8]

2.1 The fair coin is the flattest

Lemma 2.3 (termwise minimality). *For every integer $N \geq 0$ the function $f_N(q) = q^N + (1-q)^N$ on $(0,1)$ is symmetric about $q = \frac{1}{2}$ and convex, and $f_N(q) \geq f_N(\frac{1}{2}) = 2^{1-N}$, with strict inequality for $q \neq \frac{1}{2}$ as soon as $N \geq 2$.*

Proof. $f_N(1-q) = f_N(q)$ is immediate. For $N \geq 2$, $f'_N(q) = N[q^{N-1} - (1-q)^{N-1}]$ and $f''_N(q) = N(N-1)[q^{N-2} + (1-q)^{N-2}] > 0$, so f_N is strictly convex; and $t \mapsto t^{N-1}$ is strictly increasing, so $f'_N < 0$ on $(0, \frac{1}{2})$ and $f'_N > 0$ on $(\frac{1}{2}, 1)$. For $N \in \{0,1\}$, f_N is the constant 2 or 1. $\square$

Remark 2.4 (formally verified). Lemma 2.3 is machine-checked. `termwise_min` gives the inequality for every N and every $q \in [0,1]$, `termwise_min_strict` its strictness for $N \geq 2$ and $q \neq \frac{1}{2}$, and `termwise_min_iff` the equality case as a biconditional; the normalisation $2 \cdot (\frac{1}{2})^N = 2^{1-N}$ is `two_mul_half_pow`. The non-strict bound is proved twice, once from convexity of x^N on $[0,\infty)$ and once from Bernoulli's inequality (`termwise_min'`), so the two proofs check the *statement* against each other and not merely the tactic script. No `sorry`, and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`. [Lean-verified; `Pnp/Theory/TermwiseMin.lean`]

Theorem 2.5 (the fair coin minimises the invariant). *For every finite increasing sequence A of odd positive integers with $|A| \geq 2$ and every $q \in (0,1)$,*

$$\Gamma^{(q)}(A) \ \geq\ \Gamma^{(1/2)}(A),$$

with equality if and only if $q = \frac{1}{2}$.

Proof. Sum Lemma 2.3 over d. For strictness it suffices that some $d \leq D$ has $N_d \geq 2$: take any $d \geq (a_2-1)/2$, which is at most $D = (a_k-1)/2$ because $a_2 \leq a_k$. $\square$

Remark 2.6 (what the theorem says). $q(1-q)$ is the per-element variance of the coin. Biasing the coin in either direction lowers that variance and raises the invariant: **the more predictable the coin, the more rugged the landscape**. The effect is exactly symmetric in $q \leftrightarrow 1-q$, as it must be, since complementing every configuration exchanges q with $1-q$ and n with $T-n$.

Corollary 2.7 (closed form for the odd numbers). *For $A = (1,3,\ldots,2k-1)$ one has $N_d = d$ for $d \leq k$, so in the limit $k \to \infty$*

$$\Gamma^{(q)} = 1 + \frac{q}{1-q} + \frac{1-q}{q} = \frac{1}{q(1-q)} - 1 \ \geq\ 3,$$

with equality only at $q = \frac{1}{2}$: the invariant of the odd numbers is the reciprocal of the coin's variance, minus one. *[verified against the summed series to $5.3 \cdot 10^{-15}$ over $q = 0.10, \ldots, 0.80$; e5_r098]*

2.2 Verification

[experimentally confirmed] Measured $\mathrm{lm}_q/\deg_q$ at n_q against the point prediction of Definition 2.1, relative error:

profile	q	$k=60$	$k=100$	$k=140$
odds	0.20	$1.00\cdot10^{-2}$	$3.16\cdot10^{-3}$	$1.49\cdot10^{-3}$
odds	0.30	$2.63\cdot10^{-3}$	$8.02\cdot10^{-4}$	$3.75\cdot10^{-4}$
odds	0.50	$2.73\cdot10^{-4}$	$5.99\cdot10^{-5}$	$2.19\cdot10^{-5}$
primes	0.30	$2.18\cdot10^{-3}$	$5.02\cdot10^{-4}$	$2.20\cdot10^{-4}$
primes	0.50	$1.73\cdot10^{-4}$	$2.26\cdot10^{-5}$	$7.32\cdot10^{-6}$

There are no free parameters. Two features: the error decays faster than $1/k$ (fitted exponent 2.25 at $q=0.2$ to 3.0 at $q=0.5$), and it grows monotonically as q leaves $\frac{1}{2}$, by a factor ≈ 40 from $q=0.5$ to $q=0.2$ at fixed k. The second is what the biased flatness hypothesis will have to quantify. [**TODO** state the biased (H); design after §2.3.]

Remark 2.8 (the assignment, and a test that could not decide it). Definition 2.1 is symmetric in $q\leftrightarrow 1-q$, and so is the exact complementation identity $\mathrm{lm}_q(n)=\mathrm{lm}_{1-q}(T-n)$ — at the q-tilted centre, $T-qT=(1-q)T$. Neither can therefore distinguish the assignment of q^{N_d} and $(1-q)^{N_d}$ to the two strata: measured at $k=100$, the ratio at $q=0.4$ and at $q=0.6$ agrees to every printed digit. The discriminating test is per stratum, $\mathrm{over}_q/\deg_q$ against $(1-q)^{N_d}$ and $\mathrm{under}_q/\deg_q$ against q^{N_d}; measured, both ratios lie in $[0.9978, 1.0018]$ over $d\le 6$, whereas a swap would disagree by a factor of 161 at $d=6$. [experimentally confirmed; `e5_r098`]

2.3 The bias breaks the reflection symmetry

[experimentally confirmed, shape] Off the q-tilted centre the deviation $\mathrm{dev}_q=(\mathrm{lm}_q/\deg_q)/\Gamma^{(q)}-1$ behaves differently at $q=\frac{1}{2}$ and away from it, and the reason is structural. Paper 3's transfer function is even in λ because the two strata sit at $\pm\delta_d$ about a common centre — the analytic form of $\mathrm{lm}(n)=\mathrm{lm}(T-n)$. Under bias that symmetry maps q to $1-q$ rather than to itself, so the odd orders need not cancel, and the first-order coefficient is

$$L^{(q)}(A) \;=\; \sum_{d\ge1}\delta_d\Big[q^{N_d}-(1-q)^{N_d}\Big],$$

which is antisymmetric in $q\leftrightarrow 1-q$ and vanishes identically at $q=\frac{1}{2}$. Measured for the odd numbers at $k=80$: $L^{(q)}=-28.634$, -8.843, 0, $+8.843$, $+28.634$ at $q=0.3,0.4,0.5,0.6,0.7$, and for the primes -83.717, -23.677, 0, $+23.677$, $+83.717$.

The data agree. Over the same range of targets, the spread (max over min) of $|\mathrm{dev}_q-\mathrm{dev}_q(0)|$ divided by λ_q and by λ_q^2:

profile	q	divided by λ_q	divided by λ_q^2
odds	0.30	**1.329**	9.644
odds	0.50	5.091	**1.001**
primes	0.30	**1.284**	8.858

So the response is linear in λ under bias and quadratic at $q=\frac{1}{2}$, where $(\mathrm{dev}-\mathrm{dev}_0)/\lambda^2$ is constant to four digits (19.8186 to 19.8352 across a fivefold range of the target offset). [experimentally confirmed; `e6_r100`] [**TODO** the primes at $q=\frac{1}{2}$ are not a clean test in this range: $\mathrm{dev} - \mathrm{dev}_0$ crosses zero inside it, and dropping the single point where it is $2.7 \cdot 10^{-7}$ leaves the quadratic ratio in $[45.06, 48.86]$. Re-measure on a target range that does not straddle the crossing.]

3 The biased transfer function

3.1 The sign of λ, which now matters

Paper 3's transfer function is even in λ, so the sign of the slope is not observable there. Splitting the cosh makes it observable, so we fix it first:

$$\lambda_q(n) \;=\; +\frac{d}{dm}\log r_A^q(m)\Big|_{m=n} \;=\; \tfrac{1}{4}\log\frac{r_A^q(n+2)}{r_A^q(n-2)} + O(\cdot),$$

positive below the biased centre. This is the convention under which $\lambda \approx s$; paper 3's def:slope carried the opposite sign and has been corrected, with no numerical consequence there for exactly the reason just given.

3.2 The derivation

Take $P_{q,s}$: element a present with weight $q\,e^{-sa}$, absent with weight $1-q$, so $p_a = qe^{-sa}/((1-q)+qe^{-sa})$ and $p_a = q - q(1-q)sa + O(s^2a^2)$. Two consequences, both absent at $q = \frac{1}{2}$:

- the per-element variance is $a^2 p_a(1-p_a) = q(1-q)a^2(1+O(sa))$, so the layer weight of paper 3's rem:mushift becomes $e^{-\lambda^2 q(1-q)s_d/2}$;
- removing I_d removes mean $q\sigma_d$, so the two strata no longer sit at $\pm\delta_d$ about a common centre. They sit at

  $$\delta_d^+ = d + q\sigma_d, \qquad -\delta_d^-, \quad \delta_d^- = d + (1-q)\sigma_d.$$

  The bias splits the pair asymmetrically, in the ratio $q : (1-q)$.

Hence

$$\Phi^{(q)}(\lambda) \;=\; 1 + \sum_{d\ge 1}\Big[(1-q)^{N_d} e^{\lambda\delta_d^+} + q^{N_d} e^{-\lambda\delta_d^-}\Big] e^{-\lambda^2 q(1-q)s_d/2}. \tag{1}$$

[derived; three internal checks below are exact, the point prediction is verified]

Three checks are built in, and all three hold to machine zero ($0.00\cdot 10^0$, two profiles, $k = 60$): $\Phi^{(q)}(0) = \Gamma^{(q)}$ of Definition 2.1; the complementation symmetry $\Phi^{(q)}(\lambda) = \Phi^{(1-q)}(-\lambda)$, which holds because $\delta^+(q) = \delta^-(1-q)$; and $\Phi^{(1/2)}$ is paper 3's Φ term by term. [experimentally confirmed; `e10_r105`]

3.3 Tilted rigidity

Proposition 3.1 (the biased landscape responds at first order)**.** $\Phi^{(q)\prime}(0) = \sum_{d\ge 1}\Big[(1-q)^{N_d}\delta_d^+ - q^{N_d}\delta_d^-\Big]$, *which vanishes identically at $q = \frac{1}{2}$ and is non-zero otherwise. So rigidity at a biased measure is not flatness: it is a* tilt *of a definite slope, and the slope is a point prediction with no free parameter.*

This is what §2.3 demanded of the derivation, and it is a sharper demand than it looks. The naive guess — keep paper 3's $\delta_d = d + \sigma_d/2$ and merely split the weights — gives instead $\sum[(1-q)^{N_d} - q^{N_d}]\delta_d$, and the two differ by exactly $(\frac{1}{2} - q)\sum_d \sigma_d[q^{N_d} + (1-q)^{N_d}]$: zero at $q = \frac{1}{2}$, and 20–32% of the prediction at the q tested. The measurement therefore decides between them.

3.4 Verification

The estimator must kill constants, because the point prediction of §2.2 carries an $O(10^{-3})$ offset at the biased centre which is the same size as the linear term. So we use a central difference about n_q: with $n_\pm = \lfloor (q \pm hq)T \rceil$,

$$\widehat{L}(h) = \frac{(\mathrm{lm}_q/\deg_q)(n_+) - (\mathrm{lm}_q/\deg_q)(n_-)}{\lambda_q(n_+) - \lambda_q(n_-)}.$$

Ratios of $\widehat{L}$ to the two candidates, at $h = 0.01$:

profile	q	separation	$\widehat{L}/\Phi^{(q)\prime}(0)$	$\widehat{L}$/naive
odd numbers, $k = 120$	0.30	31.6%	**0.9945**	0.6805
odd numbers, $k = 120$	0.40	19.9%	**0.9976**	0.7991
odd numbers, $k = 120$	0.50	—	$\Delta(\mathrm{lm}_q/\deg_q) = 0$ exactly	
odd numbers, $k = 120$	0.60	19.9%	**0.9976**	0.7991
primes, $k = 120$	0.30	32.3%	**0.9981**	0.6754
primes, $k = 120$	0.40	20.2%	**0.9983**	0.7967

Over all 36 discriminating rows (two profiles, $k = 80, 120$, four q, four h), the median $|\widehat{L}/\Phi^{(q)\prime}(0) - 1|$ is 0.0041 against 0.2038 for the naive candidate. The one row above 10% is the primes at $k = 80$, $q = 0.3$, $h = 0.01$, where $\Delta\lambda = 8.5 \cdot 10^{-5}$ and the integer rounding of $n_\pm$ is a visible fraction of the step; the same row at $k = 120$ reads 0.9981. At $q = \frac{1}{2}$ the numerator is 0 or $4.4 \cdot 10^{-16}$, below the stated floor — the fair coin has no linear response, which is the content of Proposition 3.1 at its zero. [experimentally confirmed; e10_r105]

The split is also visible stratum by stratum, which tests the offsets directly rather than through their sum: at $k = 80$, $q = 0.3$, $\mathrm{over}_q/\deg_q$ against $(1-q)^{N_d} e^{\lambda\delta_d^+} e^{-\lambda^2 q(1-q)s_d/2}$ agrees to within 0.24% over $d = 1, \ldots, 6$, and $\mathrm{under}_q/\deg_q$ against its partner to within 0.9%. [experimentally confirmed; e10_r105]

3.5 The biased hypothesis

(H_q): $\sum_d [(1-q)^{N_d}(\delta_d^+)^2 + q^{N_d}(\delta_d^-)^2] < \infty$, uniformly for $q \in [q_0, 1-q_0]$. Since $\max(q, 1-q) \leq 1 - q_0 < 1$, the geometric damping of paper 3's (H) survives with base $1 - q_0$, and the constant degrades as $q_0 \to 0$ — which is the $\sim 40\times$ growth of the §2.2 error already measured between $q = \frac{1}{2}$ and $q = 0.2$. The $\alpha = 1$ threshold of paper 3 is unchanged, the damping argument being base-agnostic. [derived; the degradation matches the measured trend]

4 Random sequences, unconditionally

Before the theorem, the floor it has to stand on. Let A be drawn from the Cramér model on the odd integers: each odd $m \in [3, N]$ is included independently with probability $2/\log m$, and N is fixed by $\mathbb{E}|A| = k$. Nothing is conditioned, so $|A|$ fluctuates from instance to instance. That is deliberate: $kR_A(k)$ is the k-invariant quantity of paper 3's rem:onek, so an ensemble of genuinely different sizes tests the $1/k$ law for free. Twenty instances at each of two sizes, each through the whole paper-3 pipeline with its own N_d and σ_d:

$\mathbb{E}\lvert A\rvert$	range of k	(H) series	mean kR	sd	sd/mean
120	101–135	133.79–320.79	214.96	10.58	4.92%
180	146–191	132.94–471.77	206.32	6.94	3.37%

at $x = 0.06$; $c_A = kR/100$. Three things, and the third is the one to keep. [experimentally confirmed; e7_r102]

(i) The hypothesis holds on the ensemble. All forty instances have a bounded (H) series and a window that tracks the primes'. This is a precondition, not a result: a theorem about random sequences is unconditional only if the random sequences satisfy (H) themselves.

(ii) The constant concentrates. The spread across instances is under 5% and falls with k, while the mean of kR moves by 4% between the two sizes — the $1/k$ law of paper 3 surviving an ensemble it was not fitted on.

(iii) The primes sit inside the ensemble and the odd numbers do not. Against the $k \approx 180$ ensemble (2.0632 ± 0.0694 in c_A), the primes score $z = -0.39$ and the odd numbers $z = -2.70$, at every one of the four targets. In this statistic the primes are indistinguishable from a random odd sequence of the same density; the odd numbers are a different object.

There is one place where the random model is *not* like the primes, and it is the place that matters for the minor arcs. Write $h_d(q) = \max_{(r,q)=1} \left(\prod_{a \in B_d} |\cos(\pi r a/q)|\right)^{1/|B_d|}$. Since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a, the modulus-4 floor $h_d(4) = 1/\sqrt{2}$ is deterministic, not merely universal. Paper 2's vanishing lemma kills the modulus-$2m$ peak (m odd) as soon as B_d contains an element $\equiv 3 \pmod 6$. A set of primes contains exactly one such element, namely 3, and 3 leaves B_d at $d \geq 2$, so the primes keep $h_d(6) = \sqrt{3}/2$; a random odd sequence contains about $|B_d|/3$ of them and loses the peak at every d. Measured over $q \leq 120$ at $d = 1, 2, 5$: every random instance has $\max_{q \geq 3} h_d(q) = 1/\sqrt{2}$ attained at $q = 4$, and the primes have $\sqrt{3}/2$ at $q = 6$. [experimentally confirmed; e7_r102]
So the random theorem should come out with the geometric rate $1/\sqrt{2} = 0.7071$ per element rather than the $\sqrt{3}/2 = 0.8660$ of papers 1–2: *the primes are the harder case, and the modulus-*6 *ripple is a structural feature of the primes that the random model destroys.*

4.1 The modulus-4 theorem

The observation above is not an accident of the ensemble; it is a theorem about residues, and it is the exact counterpart of paper 2's modulus-6 theorem for a different equidistribution class. Paper 2 averages $|\cos|$ over the *reduced* residues, which is the right class for the primes. A random odd sequence is equidistributed over the *odd* residues instead, so the relevant quantity is a different one.

Definition 4.1. For $q \geq 2$ let $R_q = \mathbb{Z}/q$ if q is odd and $R_q = \{j : j \text{ odd}\} \subset \mathbb{Z}/q$ if q is even — for q even and m odd, $m \bmod q$ is odd, so R_q is the set of residues an odd sequence can occupy. Put

$$M_{\mathrm{odd}}(q) \;=\; \Big(\prod_{j \in R_q} |\cos(\pi j/q)|\Big)^{1/|R_q|}.$$

For r coprime to q the map $j \mapsto jr$ permutes R_q, so $M_{\mathrm{odd}}(q)$ does not depend on the numerator of the rational r/q.

Lemma 4.2 (closed form)**.** *$M_{\mathrm{odd}}(q) = 2^{1/q-1}$ for q odd. For $q = 2u$ even, $M_{\mathrm{odd}}(q) = 2^{1/u-1}$ if u is even, and $M_{\mathrm{odd}}(q) = 0$ if u is odd.*

Proof. Write $|\cos(\pi j/q)| = |1 + e^{2\pi i j/q}|/2$.

If q is odd, $e^{2\pi i j/q}$ runs over all q-th roots of unity as j runs over R_q. Setting $x = -1$ in $x^q - 1 = \prod_\rho (x - \rho)$ gives $\prod_\rho(-1-\rho) = -2$, and $\prod_\rho(-1-\rho) = (-1)^q \prod_\rho (1+\rho) = -\prod_\rho(1+\rho)$ because q is odd, so $\prod_\rho(1+\rho) = 2$ and $\prod_j |\cos(\pi j/q)| = 2^{-q} \cdot 2 = 2^{1-q}$.

If $q = 2u$ and j is odd, $e^{2\pi i j/q} = e^{\pi i j/u}$ runs over the u roots of $x^u + 1$. Setting $x = -1$ in $x^u + 1 = \prod_\rho(x - \rho)$ gives $(-1)^u + 1 = (-1)^u \prod_\rho(1+\rho)$, so $\prod_\rho(1+\rho) = 1 + (-1)^u$, which is 2 for u even and 0 for u odd. Hence $\prod_j |\cos| = 2^{-u} \cdot 2 = 2^{1-u}$ or 0. Taking $|R_q|$-th roots gives the statement. ∎

Theorem 4.3 (the modulus-4 theorem)**.** $\max_{q\geq 2} M_{\mathrm{odd}}(q) = \dfrac{1}{\sqrt{2}}$, *attained at $q = 4$ and nowhere else.*

Proof. $v \mapsto 2^{1/v-1}$ is strictly decreasing. For q odd, $q \geq 3$, Lemma 4.2 gives $M_{\mathrm{odd}}(q) = 2^{1/q-1} \leq 2^{-2/3}$. For $q \equiv 2 \pmod 4$ it gives 0. For $q \equiv 0 \pmod 4$, write $q = 2u$ with u even, $u \geq 2$; then $M_{\mathrm{odd}}(q) = 2^{1/u-1} \leq 2^{-1/2}$ with equality exactly when $u = 2$, i.e. $q = 4$. And $2^{-2/3} < 2^{-1/2}$. □

Verified against the product for all $q \leq 200$, worst discrepancy $4.4 \cdot 10^{-16}$. [proved; and checked numerically, `mq4_r107`]

Remark 4.4 (what is machine-checked here, and what is not). The two halves of the argument have different statuses and the difference is worth stating. *The maximisation is formally verified.* Taking the closed form of Lemma 4.2 as a definition, `ModdCF_le` gives $M_{\mathrm{odd}}(q) \leq 2^{-1/2}$ for every $q \geq 2$, `ModdCF_eq_iff` the equality case as a biconditional, `ModdCF_lt_of_ne_four` the strict bound off $q = 4$ by a second route, and `two_rpow_neg_half` the normalisation $2^{-1/2} = 1/\sqrt{2}$. This is the step where an error would hide — a case split over three residue classes and the comparison of $2^{-2/3}$ with $2^{-1/2}$ — and it is now checked by machine. *The closed form itself is not formalised*: it rests on two evaluations of a root-of-unity product, is proved above, and is checked against the product for every $q \leq 200$. [Lean-verified (maximisation only); `Pnp/Theory/ModFour.lean`]

Remark 4.5 (the two classes, side by side).

	class of residues	extremal q	rate per element
primes (paper 2)	reduced	6	$\sqrt{3}/2 = 0.8660$
random odd A (here)	odd	4	$1/\sqrt{2} = 0.7071$

The two disagree exactly at $q \equiv 2 \pmod 4$: $M(6) = \sqrt{3}/2$ but $M_{\mathrm{odd}}(6) = 0$, because $j = 3$ is an odd residue mod 6 and $\cos(\pi/2) = 0$. A set of primes contains at most one element $\equiv 3 \pmod 6$, and it leaves B_d at $d \geq 2$; a random odd sequence contains a positive proportion of them at every d. *So the primes are the harder case, by a factor $\sqrt{3/2} = 1.2247$ in the geometric rate*, and paper 2's $\sqrt{3}/2$ is a feature of the primes rather than a limitation of the method.

4.2 The rate, and what is still missing

Proposition 4.6 (minor-arc rate for random odd sequences)**.** *Let A be drawn from the Cramér model above and let $F_A(\theta) = \prod_{a\in A} |\cos(\pi a\theta)|$. Then almost surely*

$$\lim_{b\to\infty} \max_{\theta \text{ minor}} F_A(\theta)^{1/b} = \frac{1}{\sqrt{2}},$$

the maximum being attained in a shrinking neighbourhood of $\theta = \frac{1}{4}$.

The arithmetic half is Theorem 4.3. Here is the rest, in the four steps it actually takes; each is elementary, and the point of the section is that none of them needs Siegel–Walfisz, Kumchev, or any ineffective constant.

Write $X_m(\theta) = -\log|\cos(\pi m\theta)| \in [0, \infty]$, so that $-\log F_A(\theta) = \sum_m \xi_m X_m(\theta)$ with ξ_m the independent indicators of the model. Only a *lower* bound on this sum is wanted, which is what makes the argument easy.

Step 1 (truncate; the unbounded tail is free). Put $X_m^{(M)} = \min(X_m, M)$. Then $\sum \xi_m X_m \geq \sum \xi_m X_m^{(M)}$, so it suffices to bound the truncated sum below, and the summands now lie in

$[0, M]$. At $\theta = \frac{1}{4}$ truncation is inactive once $M > \frac{1}{2}\log 2$, since $X_m(\frac{1}{4}) = \frac{1}{2}\log 2$ for every odd m.

Step 2 (the mean, and where Theorem 4.3 enters). For $\theta = r/q$ the values $X_m(\theta)$ over odd m are a permutation of $\{-\log|\cos(\pi j/q)| : j \in R_q\}$ repeated, so $\mathbb{E}\sum_m \xi_m X_m(r/q) = (-\log M_{\text{odd}}(q) + o(1))\mathbb{E}|A|$, and Theorem 4.3 says this is at least $\frac{1}{2}\log 2$ per element, with a strict gap $\delta(q) > 0$ for every $q \neq 4$ and $\delta(q) \to \log 2$ as $q \to \infty$. *This is the only step where arithmetic appears*, and it is the step where the primes and a random odd sequence part company. For general θ one reduces to a nearby rational by Dirichlet approximation; the arc geometry is paper 2's and is used verbatim, only the peak-height input changing from M to M_{odd}.

Step 3 (concentration). The summands are independent and lie in $[0, M]$, so Hoeffding gives $\Pr[\sum_m \xi_m X_m^{(M)} < \mathbb{E} - t] \leq \exp(-2t^2/(\pi(N)M^2))$, and with $t = \frac{1}{2}\delta b$ this is $\exp(-c\delta^2 b/M^2)$. No second moment of an exponential sum is needed — the randomness is in the *sequence*, not in the phases.

Step 4 (net, then Borel–Cantelli). On the region where $X_m < M$ one has $|\cos(\pi m\theta)| \geq e^{-M}$, hence $|\partial_\theta X_m^{(M)}| = \pi m|\tan(\pi m\theta)| \leq \pi N e^M$, so a net of spacing $\eta = \varepsilon/(\pi b N e^M)$ controls θ, and the net has $O(bNe^M/\varepsilon)$ points. The union bound survives as long as $c\delta^2 b/M^2 \gg \log N + M$, and $b \asymp N/\log N$ in the model, so any $M = M(b) \to \infty$ with $M = o(b^{1/3})$ will do. Summability over b then gives the almost-sure statement by Borel–Cantelli. [proof skeleton — Theorem 4.3 proved, Steps 1–4 as above with the constants left implicit; measured in the two tables of this section] [**TODO** write Steps 2–4 out with explicit constants. The only non-routine point is the Dirichlet reduction in Step 2, and paper 2's §major/minor decomposition already contains it; what has to be checked is that its arc radii are still admissible when the peak height drops from $\sqrt{3}/2$ to $1/\sqrt{2}$ — they should only get easier.]

Measured, at $\theta = r/q$ for $q \leq 12$ and eight instances at $b \approx 190$: the mean of $b^{-1}\sum_a -\log|\cos(\pi a/q)|$ agrees with $-\log M_{\text{odd}}(q)$ within the $O(b^{-1/2})$ fluctuation at every q, and at $q = 4$ it is 0.346574 with standard deviation 0.000000 — deterministic, as Definition 4.1 says it must be. [experimentally confirmed; `mq4_r107`]

Remark 4.7 ($\theta = \frac{1}{4}$ is the limit of the maxima, not a maximum). A first scan of $F_A(\theta)^{1/b}$ over a fine net asked whether anything exceeds $1/\sqrt{2}$, and found that something does. That is not a counterexample and the reason is worth stating, because it fixes the form of Proposition 4.6. Moving off $\frac{1}{4}$ by δ changes $\log F_A^{1/b}$ by $\delta\, b^{-1}\sum_a(\pm\pi a) + O(\delta^2)$, and the signs depend on a mod 4, so the first-order term is a random walk of size $b^{-1}\sqrt{S_2}$. Optimising δ gives an excess of $O(1/b)$ in the exponent — a *bounded multiplicative factor* on F_A, not a geometric one. Measured over $b = 111, 190, 310, 542$: $\max_\theta F_A^{1/b} - 1/\sqrt{2}$ is $9.0 \cdot 10^{-4}$, $1.8 \cdot 10^{-3}$, $1.1 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and $b\log(\sqrt{2}\max_\theta F_A^{1/b})$ stays in $[0.05, 0.89]$ — $O(1)$, as derived. The argmax sits at $\theta = 0.24996, \ldots, 0.25001$. [experimentally confirmed; `mq4_r107`]

5 Two questions inherited from paper 3

(i) The c_d fold-in is settled, and it goes paper 3's way. The share of the residual that c_d carries falls like $k^{-2\alpha}$ — measured exponents 1.98, 2.35, 2.99 against predicted 2.00, ≈2.4, 3.00 on the odd numbers, the primes and $\alpha = \frac{3}{2}$, with the squares already below the numerical floor. So Φ as printed is right for every larger k and c_d belongs to E_k permanently; paper 3's rem:cd now says so. [experimentally confirmed, four profiles, $k \leq 300$; `e8_r103`]

(ii) $c_A = c'_A$ is still open, but sharper. Carried to $k = 300$ the ratio reaches 1.0064 (odd numbers), 1.0001 (primes), 0.9984 ($\alpha = \frac{3}{2}$), and the gap closes like $k^{-2.4}$ — faster than the $1/k$ law both constants sit on. [experimentally confirmed; `e8_r103`] [**TODO** the mechanism is

still missing, and a measured rate of convergence is evidence about a limit, not a proof of one. F45 stands: substituting s for λ makes the residual three times worse.]

6 Where this sits, and the algorithmic reading

6.1 Related work

Subset sums and number partitioning have a large landscape literature, and it is worth saying precisely where the present series does and does not meet it. That literature is overwhelmingly about *random instances*: the phase transition and barrier-tree statistics of the partitioning landscape, the local-REM description of the near-optimal spectrum, and more recently overlap-gap obstructions for local algorithms. Its objects are the energy spectrum and the dynamics; its randomness is in the instance; and its answers are laws for a typical instance.

Papers 1–3 do something different in kind: they fix the sequence and compute an *arithmetic invariant of that sequence*, $\Gamma(A) = \sum_j a_j 2^{-j}$, which the ratio $\mathrm{lm}/\deg$ converges to for *every* target. Nothing is averaged over instances, and the answer is a number one reads off A. The present paper's §4 is the one place where the two meet — there the sequence is random — and even there the question is the ratio's limit rather than the spectrum's law.

One vocabulary warning, since the words collide. "Biased Bernoulli measure" is established terminology in the theory of Bernoulli convolutions, where it denotes a self-similar measure on $\mathbb{R}$; that is a different object from P_q here, and we do not claim the term. Within the subset-sum landscape literature we have not found the biased deformation $\Gamma^{(q)}$ of §2, nor the odd-residue extremal quantity M_{odd} of §4.1, under any name. [literature pass, not a proof of novelty: absence of evidence over the searches made]

6.2 The algorithmic reading

This programme began with local search on subset-sum instances, and the invariant answers a question one asks there. If a local-search procedure ends at a uniformly chosen strict local minimum, the expected number of attempts before landing on a ground state is $\mathrm{lm}/\deg$, and papers 1–3 say that number converges to $\Gamma(A)$ — *computable from A alone, without solving a single instance, and independent of the target*. For the odd numbers it is 3; for the primes, 5.3492…

Theorem 2.5 then says something with an algorithmic flavour and a proof: *no Bernoulli bias helps*. Sampling configurations with a biased coin instead of a fair one multiplies the ratio by $\Gamma^{(q)}/\Gamma \geq 1$, strictly for $q \neq \frac{1}{2}$. A whole natural one-parameter family of restart heuristics therefore cannot beat the uniform one on this statistic, and §3 adds that the biased landscape also acquires a *first-order* sensitivity to the target that the fair one does not have — it is worse in two ways at once.

Two caveats, and they are the reason this is an outlook and not a theorem about algorithms. A count of local minima is not a distribution over basins of attraction: real local search does not terminate uniformly, and the ratio bounds the wrong quantity if the basins are very unequal. And nothing here addresses dynamics — barrier heights, escape times, or the behaviour of any particular heuristic. What the invariant does supply is a target-independent constant that any such analysis has to be consistent with, and a proved statement that one obvious way to try to improve it does not. [**TODO** if this section is ever to make an algorithmic claim, the missing object is the basin-size distribution, not more landscape statistics.]

7 Honest scope

- *Lean-verified*: Lemma 2.3 in all three forms (Remark 2.4), including the equality case; and the maximisation step of Theorem 4.3 (Remark 4.4), the closed form being taken as a definition there. The canon also passes an independent kernel replay (`lean4checker`, 12 modules, with a negative control).
- *Proved*: Lemma 2.3, Theorem 2.5, Corollary 2.7; and Remark 2.2's derivation of Definition 2.1 modulo the biased flatness hypothesis, which is not yet stated.
- *Experimentally confirmed, with the range stated*: the point prediction (§2.2, two profiles, five q, three k), the stratum assignment (Remark 2.8), the linear response under bias (§2.3).
- *Derived and verified*: $\Phi^{(q)}$ of (1) and Proposition 3.1 — three exact internal checks, and the first-order point prediction confirmed against the naive alternative on 36 discriminating rows (§3.4); the random-ensemble floor of §4; the c_d verdict of §5.
- *Proved*: Theorem 4.3 and Lemma 4.2, the arithmetic half of §4.
- *Proof skeleton*: Proposition 4.6, whose four steps are written out with the constants left implicit.
- *Not yet written*: the passage from (H_q) to a rigidity statement, and the explicit constants of Steps 2–4 in §4.

References

[1] OpenAI, *Ten advances in mathematics and theoretical computer science*, 2026, with Lean 4 certificates at `github.com/openai/ten-proofs`. Cited here only as a precedent for the genre; no mathematical result of that work is used.

[2] K. Amauchi, *The gap series of an integer sequence*. Manuscript, 2026.

[3] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes*. Manuscript, 2026.

[4] K. Amauchi, *The transfer function of subset-sum landscapes*. Manuscript, 2026.